Cognitive AI Project — Current Status Summary

What We Have Accomplished

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1. Project Structure Setup

✓ GitHub repository created

✓ Folder structure created

dataset/
ml/
models/
nlp/
simulation/
ui/
requirements.txt
README.md

✓ Team workflow setup started

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2. Dataset Creation

✓ Created dataset for decision support system

✓ Multiple categories included:

- Study First
- Job First
- Higher Studies
- Balanced Plan
- Counseling Needed

✓ Added:

- single problem cases
- double problem cases
- mixed multi-query cases

✓ Removed major label leakage issues

✓ Improved realistic prediction testing

✓ Final usable dataset prepared

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3. ML Model Training (Member 1)

✓ Used:

- TF-IDF Vectorizer
- Logistic Regression

✓ Added:

- stratify=y
- class_weight="balanced"
- better train-test split

✓ Model training completed

✓ Saved trained model:

models/model.pkl

✓ Created prediction testing file:

ml/predict.py

✓ Tested with real unseen inputs

✓ Found strengths + weak predictions

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4. NLP Module (Member 2)

preprocess.py

✓ Lowercase conversion

✓ Punctuation removal

✓ Stopword removal

✓ Tokenization

✓ Lemmatization

✓ Keyword extraction

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emotion_detection.py

✓ Emotion detection

- Stress
- Anxiety
- Confusion
- Depression
- Motivation

✓ Sentiment analysis

- Positive
- Negative
- Neutral

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5. Simulation Engine (Member 3)

simulation.py

✓ Future outcome simulation

Includes:

- short-term outcome
- long-term outcome
- risks
- benefits

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planner.py

✓ Weekly action planner

Includes:

- Week 1 to Week 4 practical roadmap

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6. Streamlit UI (Member 4)

✓ Streamlit installed

✓ UI created:

ui/app.py

✓ Connected:

- NLP
- Emotion detection
- ML prediction
- Simulation engine
- Weekly planner

✓ Full working system launched successfully

✓ Browser-based working interface completed

What We Need To Do Next

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1. Improve ML Predictions (HIGH PRIORITY)

Current issue:

Some predictions are slightly weak

Example:

low money + higher studies
→ predicted: Study First

Should be better like:

- Balanced Plan
- Job First
- Higher Studies

Need:

✓ Add targeted dataset rows

✓ Improve difficult edge cases

✓ Retrain model

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2. UI Polishing (HIGH PRIORITY)

Need:

✓ better design

✓ better spacing

✓ better colors

✓ better layout

✓ more professional look

✓ stronger presentation quality

This is important for faculty impression

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3. GitHub Final Push

✓ clean project files

✓ proper README

✓ final commit

✓ team collaboration ready

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4. Free Deployment

Deploy project on:

Recommended:

Hugging Face Spaces

or

Streamlit Community Cloud

This makes project accessible online

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5. Final PPT + Report

Need:

✓ architecture diagram

✓ workflow explanation

✓ module explanation

✓ screenshots

✓ results

✓ future scope

✓ deployment proof

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6. Viva Preparation

Need strong answers for:

✓ Why Logistic Regression?

✓ Why not FLAN-T5 first?

✓ Why train-test accuracy was high?

✓ How unseen testing works?

✓ Future scope with FLAN-T5

Final Project Status

Current Progress:

~25–30% Complete ✅

Only polishing + deployment left

Core system is already built